Chapter-5
Pulse Modulation and Demodulation

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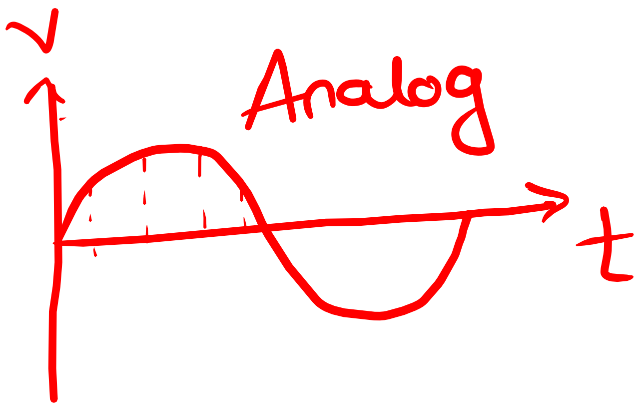
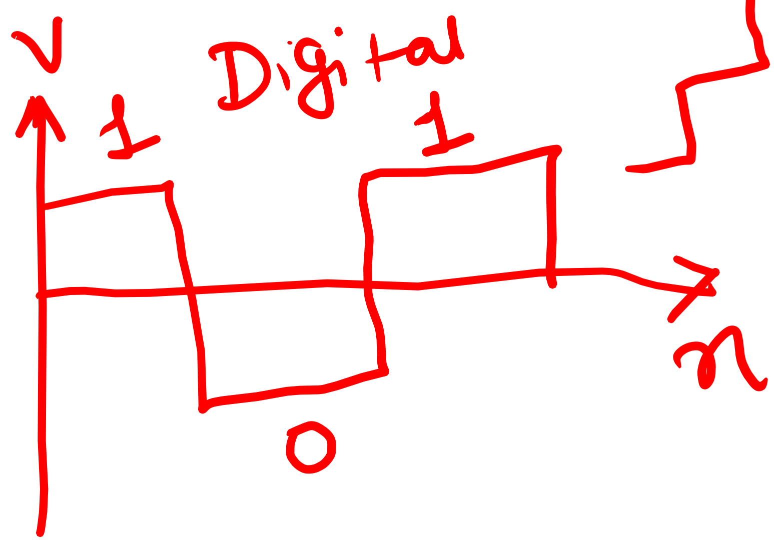
Analog to Digital Conversion

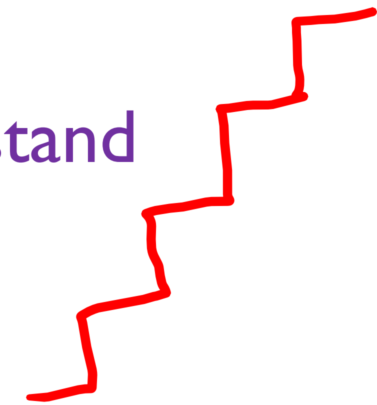
Introduction

- Your voice is analog- how does your phone send it on WatsApp?

- Why cant computers understand continuous signals?







Objective to State

- Today we will learn how real-world analog signals become digital data using Analog to Digital Conversion

3 Step Process

- Step-1 Sampling
- Step-2 Quantization
- Step-3 Encoding

Step-1 Sampling

- Sampling $\rightarrow f_s \times f_m$

- Nyquist Theorem

$$f_s \geq 2 f_m$$

maximum signal frequency

$$f_s = 2 f_m \text{ (Nyquist sampling)}$$

Step-1 Sampling

- Aliasing ($F_s < 2F_m$)

Distortion in received signal

- If signal frequency is 5 KHz , what is minimum sampling rate?

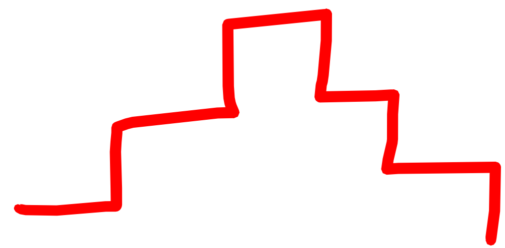
$$F_s = 2 \times 5\,KHz$$

$$F_s = 10\,KHz$$

Step-2 Quantization & Encoding

- Quantization (infinite to finite)

Quantization error



- Step Size (Δ) $= \dfrac{V_{max} - V_{min}}{L}$

- $L = 2^n$ (number of levels)

Step-3 Encoding

- Assigning Binary Levels